

Total marks (120)
Attempt questions 1 – 10
All questions are of equal value

Answer each question in a SEPARATE Writing Booklet. Extra Writing Booklets are available.

Question 1	(12 marks) Use a SEPARATE Writing Booklet.	Marks
(a)	Solve $1 - 2x < 9$.	2
(b)	Factorise $6x^2 - x - 1$.	2
(c)	Solve $ 2x - 5 = 8$.	2
(d)	Sketch the graph of $3y + x = 6$, showing the intercepts on both axes.	2
(e)	Rationalise the denominator of $\frac{5}{\sqrt{3} + 3}$.	2
(f)	Sketch $f(x) = \sqrt{9 - x^2}$.	2

Question 2 (12 marks) Use a SEPARATE Writing Booklet.**Marks**(a) Differentiate with respect to x :

(i) $(3 + x^2)^{12}$. 2

(ii) $\frac{\ln x}{e^x}$. 2

(iii) $x^2 \cos \frac{x}{2}$. 2

(b) (i) Find $\int \frac{x}{x^2 - 4} dx$. 1

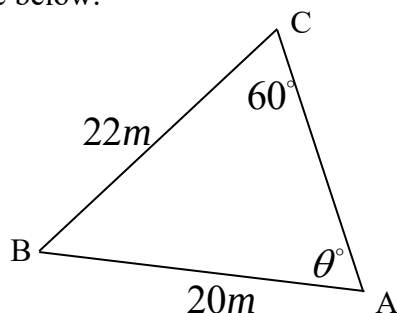
(ii) Evaluate $\int_1^4 \left(\frac{1}{x^2} - \sqrt{x} \right) dx$. 3

(iii) Evaluate $\int_0^{\frac{\pi}{8}} \sec^2 2x \, dx$. 2

Question 3 (12 marks) Use a SEPARATE Writing Booklet.

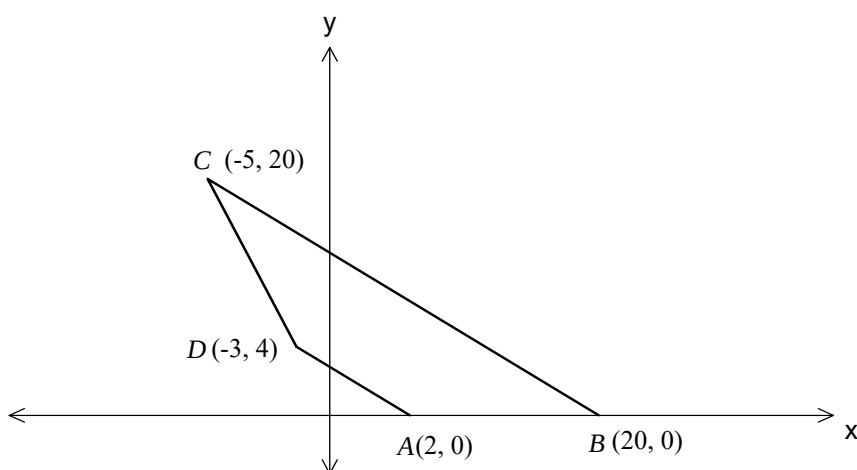
Marks

- (a) Consider the triangle below.



- | | | |
|------|--|---|
| (i) | Find the size of the angle θ , correct to the nearest degree. | 2 |
| (ii) | Find the area of the triangle, using your approximation for the angle found in (i). (correct to the nearest cm^2 .) | 1 |

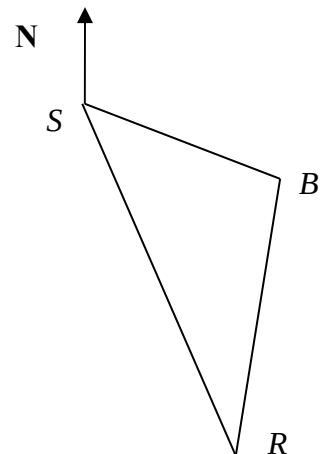
- (b)



- | | | |
|-------|--|---|
| (i) | Show that ABCD is a trapezium, by showing that AD is parallel to BC. | 2 |
| (ii) | Show that the equation of the line BC is $4x + 5y - 80 = 0$. | 2 |
| (iii) | Find the length of BC. (Leave in exact form) | 1 |
| (iv) | Find the perpendicular distance of the point D from the line BC. | 2 |
| (v) | Hence, or otherwise, find the area of the trapezium ABCD. | 2 |

Question 4 (12 marks) Use a SEPARATE Writing Booklet.**Marks**

- (a) A triathlon course begins with a 500 m swim on a bearing of 110° from the start S . This is followed by a 1800 m cycling leg on a bearing of 185° . The triathlon is completed with a run back to S .



- | | | |
|-------|---|----------|
| (i) | Copy and complete this diagram. Find the size of $\angle SBR$. | 1 |
| (ii) | How far was the run home? (nearest m) | 2 |
| (iii) | Find the bearing of S from R . (nearest degree) | 2 |
- (b) Sarah plays computer games competitively. From past experience, she has a 0.9 chance of winning a game of *Staplestory* and a 0.6 chance of winning a game of *Bota*. In one afternoon of competition she plays two games of *Staplestory* and one of *Bota*.
- | | | |
|------|---|----------|
| (i) | What is the probability that she will win all three games? | 1 |
| (ii) | What is the probability that she wins at least one game of <i>Staplestory</i> and loses the game of <i>Bota</i> ? | 2 |
- (c) Each day a runner trains for a 10 km race. On the first day she runs 1000 m, and then increases the distance by 250 m on each subsequent day.
- | | | |
|------|--|----------|
| (i) | On which day does she run a distance of 10 km in training? | 2 |
| (ii) | What is the total distance she will have run in training by the end of that day? | 2 |

Question 5 (12 marks) Use a SEPARATE Writing Booklet.

Marks

(a) Solve the equation $\sin \theta = -\cos \theta$ for $0^\circ \leq \theta \leq 360^\circ$. ($\cos \theta \neq 0$) **3**

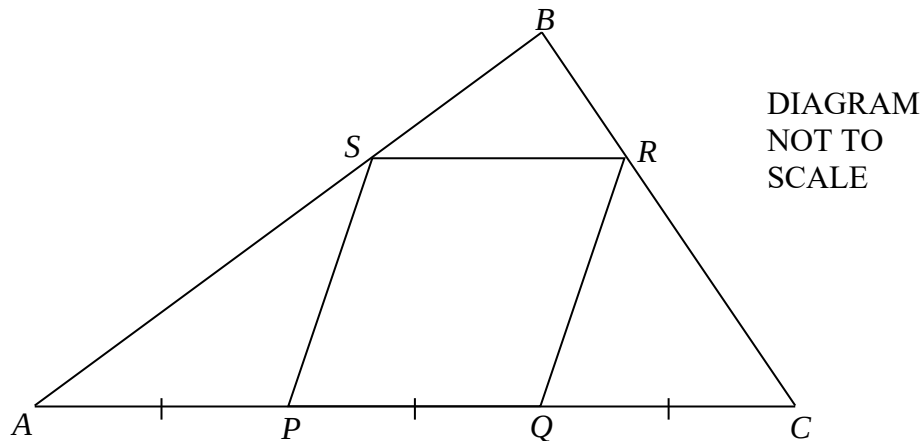
(b) If α and β are the roots of the equation $5x^2 - 3x - 2 = 0$, find the values of:

(i) $\alpha + \beta$ **1**

(ii) $\alpha\beta$ **1**

(iii) $\alpha^2 + \beta^2$ **2**

(c)



The diagram above shows $\triangle ABC$, where $AP = PQ = QC$ and $PQRS$ is a rhombus.

(i) If $\angle SAP = x^\circ$ prove that $\angle SPQ = 2x^\circ$ **2**

(ii) Prove that $\angle ABC = 90^\circ$ **3**

Question 6 (12 marks) Use a SEPARATE Writing Booklet.**Marks**

- (a) Consider the equation $4x^2 + kx + 1 = 0$. For what values of k does this equation have two real and distinct roots?

3

- (b) One model for the number of individuals connected to the internet worldwide is the exponential growth model.

$$N = Ae^{kt}$$

where N is the estimate for the number of individuals connected to the internet (in millions), and t is the time in years after 1 January 2011. It is estimated that at the start of 2012, when $t = 1$, there will be 1800 million individuals connected to the internet, while at the start of 2013, when $t = 2$, there will be 2400 million individuals connected to the internet.

- (i) Show that $A = 1350$ and $k = \ln\left(\frac{4}{3}\right)$. **2**

- (ii) According to the model, during which month and year will the number of individuals connected to the internet first exceed 4000 million. **2**

- (iii) At what rate will the number of individuals be increasing in 2015? **2**

- (c) Use Simpson's Rule with five function values to estimate the area under the curve $y = \ln(x+1)$ and the x -axis between $x = 1$ and $x = 5$. (correct to two decimal places)

3

Question 7 (12 marks) Use a SEPARATE Writing Booklet.**Marks**

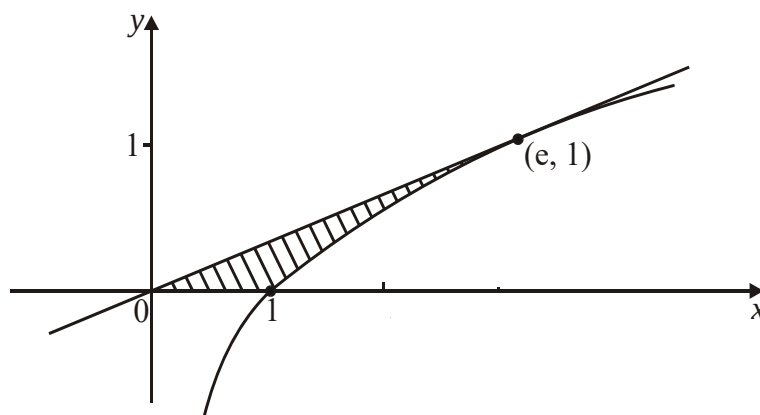
(a) Evaluate $\sum_{n=1}^{\infty} 5\left(\frac{1}{2}\right)^n$. 2

(b) Let $f(x) = \frac{x^3}{3} + x - 3$

(i) Show that the graph of $y = f(x)$ has no stationary points. 2

(ii) For what values of x is the graph of $y = f(x)$ concave up? 1

(c)



(i) Find the equation of the tangent line to the curve $y = \ln x$ at the point $(e, 1)$, and verify that the origin is on this line. 2

(ii) Show that $\frac{d}{dx}(x \ln x - x) = \ln x$. 2

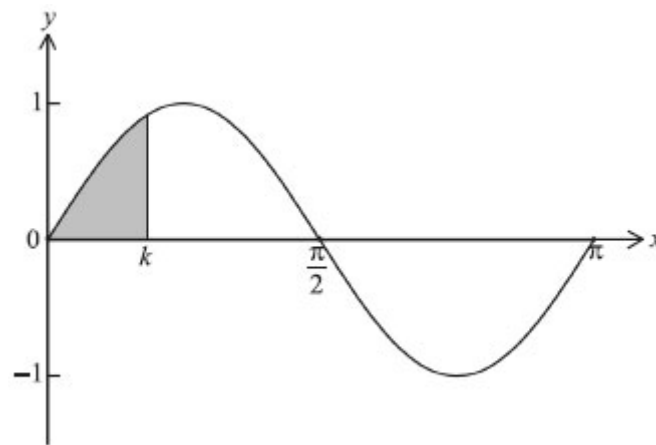
(iii) The diagram shows the region enclosed by the curve $y = \ln x$, the tangent line and the line $y = 0$. Use the result of part (ii) to show that the area of this region is $\frac{e}{2} - 1$. 3

Question 8 (12 marks) Use a SEPARATE Writing Booklet.**Marks**

- (a) The velocity, v , in m s^{-1} of a particle moving in a straight line is given by $v = e^{3t-2}$, where t is the time in seconds.

- (i) Find the acceleration of the particle at $t=1$. 1
- (ii) At what value of t does the particle have a velocity of 22.3 m s^{-1} ? (correct to one decimal place) 2
- (iii) Find the distance travelled in the first second. (correct to three decimal places) 2

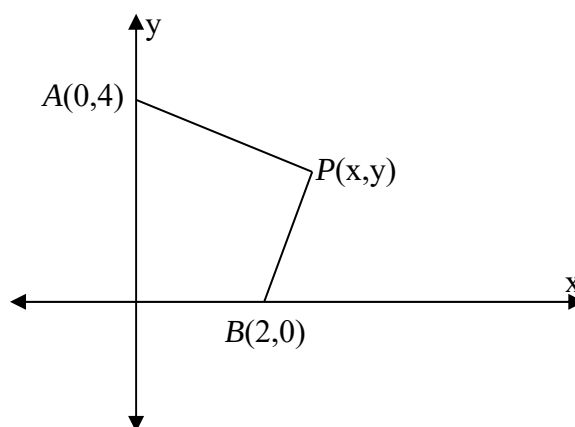
- (b) The graph of $y = \sin 2x$ from $0 \leq x \leq \pi$ is shown below.



The area of the shaded region is 0.85. Find the value of k . (correct to two decimal places)

3

- (c) A point $P(x,y)$ moves in such a way that PA is perpendicular to PB .



- (i) Show that the equation of the locus is given by $x^2 - 2x + y^2 - 4y = 0$. 2
- (ii) Find the centre and radius of this circle. 2

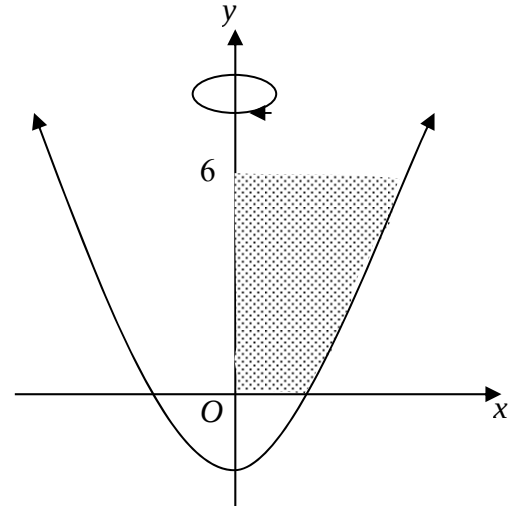
Question 9 (12 marks) Use a SEPARATE Writing Booklet.**Marks**

- (a) The diagram shows the region bounded by the curve $y = 3x^2 - 12$,

4

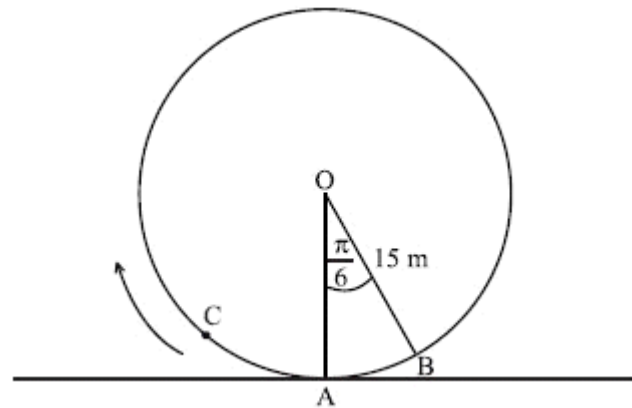
the line $y = 6$, and the x and y axes.

Find the exact volume of the solid of revolution formed when the shaded region is rotated about the y axis.



- (b) A Ferris wheel with centre O and a radius of 15 metres is represented in the diagram below. Initially seat A is at ground level.

The next seat is B , where $\angle AOB = \frac{\pi}{6}$.



- (i) Find the length of the arc AB . **1**
- (ii) Find the area of the sector AOB . **1**
- (iii) The wheel turns clockwise through an angle of $\frac{2\pi}{3}$. Find the height of A above the ground. **3**
- (iv) The height, h metres, of seat C above the ground after t minutes, can be modelled by the function

$$h(t) = 15 - 15 \cos \left(2t + \frac{\pi}{4} \right).$$
 3

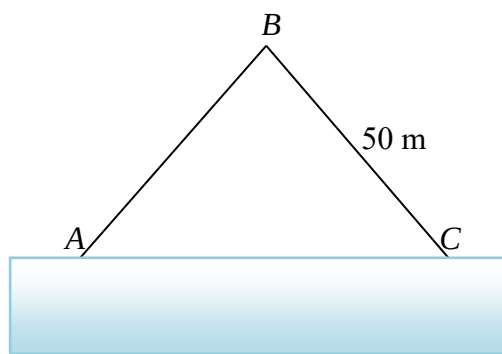
Find the time at which the height is changing most rapidly.

Question 10 (12 marks) Use a SEPARATE Writing Booklet.**Marks**

- (a) A car dealership has a car for sale for a cash price of \$35 000. It can also be bought on terms over five years. The first six months are interest free and after that interest is charged at the rate of 1.5% per month on the balance owing for that month. Repayments are to be made in equal monthly installments of \$ M with the first repayment applied at the end of the first month. A customer agrees to buy the car on these terms.

Let \$ A_n be the amount owing at the end of the n th month.

- (i) Find an expression for A_6 . 1
 - (ii) Show that $A_8 = (35\,000 - 6M)1.015^2 - M(1 + 1.015)$. 2
 - (iii) Find an expression for A_{60} . 2
 - (iv) Find the value of M . 2
- (b) An isosceles triangular pen is enclosed by two fences AB and BC each of length 50 m, and a river is the third side.



- (i) If $AC = 2x$ m, show that the area of the triangle is given by:

$$A(x) = x\sqrt{2500 - x^2}. \quad \text{2}$$

- (ii) Hence find the length of AC when the area is a maximum.
(correct to the nearest m)

3**END OF PAPER**

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STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x$, $x > 0$

Question 1

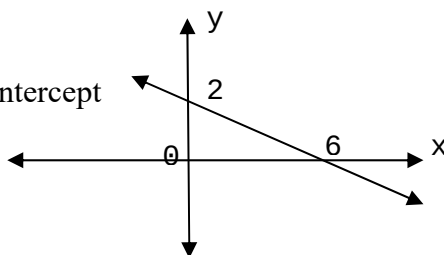
(a) $1 - 2x < 9$
 $-2x < 8$ ☒
 $x > -4$ ☒

(b) $6x^2 - x - 1$
 $= (2x - 1)(3x + 1)$ ☒☒

(c) $|2x - 5| = 8$

$2x - 5 = 8$ ☒ or $-2x + 5 = 8$ ☒
 $x = \frac{13}{2}$ $x = -\frac{3}{2}$

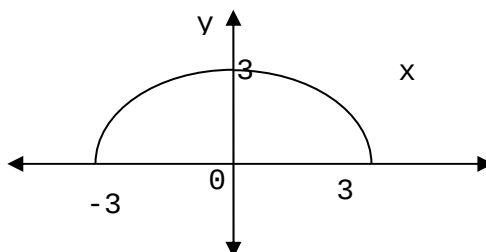
(d) ☒ ☒ One for each intercept



(e) $\frac{5}{\sqrt{3}+3} \times \frac{\sqrt{3}-3}{\sqrt{3}-3}$ ☒
 $= \frac{5\sqrt{3}-15}{3-9}$
 $= \frac{15-5\sqrt{3}}{6}$ ☒

(f)

- ☒ -A semi circle!!
☒ -Intercepts



Question 2

(a) (i) Let $y = (3 + x^2)^{12}$

$$\therefore \frac{dy}{dx} = 12(3 + x^2)^{11} \times 2x \quad \boxed{\checkmark}$$

$$\therefore \frac{dy}{dx} = 24x(3 + x^2)^{11} \quad \boxed{\checkmark}$$

(ii) Let $y = \frac{\ln x}{e^x}$

$$\therefore \frac{dy}{dx} = \frac{e^x \cdot \frac{1}{x} - e^x \ln x}{e^{2x}} \quad \boxed{\checkmark}$$

$$\therefore \frac{dy}{dx} = \frac{1 - x \ln x}{xe^x} \quad \boxed{\checkmark}$$

(iii) Let $y = x^2 \cos \frac{x}{2}$

$$\therefore \frac{dy}{dx} = -x^2 \sin\left(\frac{x}{2}\right) \times \frac{1}{2} + 2x \cos \frac{x}{2} \quad \boxed{\checkmark}$$

$$\therefore \frac{dy}{dx} = \frac{-x^2}{2} \sin\left(\frac{x}{2}\right) + 2x \cos \frac{x}{2} \quad \boxed{\checkmark}$$

(b) (i)
$$\int \frac{x}{x^2 - 4} dx = \frac{1}{2} \int \frac{2x}{x^2 - 4} dx$$
$$= \frac{1}{2} \ln(x^2 - 4) + c \quad \boxed{\checkmark}$$

(ii)
$$\int_1^4 \left(\frac{1}{x^2} - \sqrt{x} \right) dx = \int_1^4 \left(x^{-2} - x^{\frac{1}{2}} \right) dx \quad \boxed{\checkmark}$$

$$= \left[\frac{1}{-x} - \frac{2x^{\frac{3}{2}}}{3} \right]_1^4 \quad \boxed{\checkmark}$$

$$= \left(\frac{1}{-4} - \frac{2(4)^{\frac{3}{2}}}{3} \right) - \left(\frac{1}{-1} - \frac{2(1)^{\frac{3}{2}}}{3} \right)$$

$$= -3\frac{11}{12} \quad \boxed{\checkmark}$$

(iii)
$$\int_0^{\frac{\pi}{8}} \sec^2 2x dx = \frac{1}{2} [\tan 2x]_0^{\frac{\pi}{8}} \quad \boxed{\checkmark}$$

$$= \frac{1}{2} \tan \frac{\pi}{4} - \frac{1}{2} \tan 0$$

$$= \frac{1}{2} \quad \boxed{\checkmark}$$

Question 3

$$(a) (i) \frac{\sin \theta}{22} = \frac{\sin 60}{20} \quad \boxed{\checkmark}$$

$$\sin \theta = \frac{22 \sin 60}{20}$$

$$\therefore \theta = 72 \text{ (nearest degree)} \quad \boxed{\checkmark}$$

$$(ii) \text{ Area} = \frac{1}{2} \times 20 \times 22 \times \sin 48$$

$$= 163.49186 \text{ m}^2 \text{ (nearest cm}^2\text{)} \quad \boxed{\checkmark}$$

$$(b) (i) \text{ Gradient } AD = \frac{4}{-3-2} = \frac{-4}{5}$$

$$\text{Gradient } BC = \frac{20}{-5-20} = \frac{-4}{5}$$

$$\therefore AD \parallel BC \quad \boxed{\checkmark} \boxed{\checkmark}$$

$$(ii) \text{ Equation of } BC: y - y_1 = m(x - x_1)$$

$$y - 0 = \frac{-4}{5}(x - 20) \quad \boxed{\checkmark}$$

$$\therefore 5y = -4x + 80 \quad \boxed{\checkmark}$$

$$\therefore 4x + 5y - 80 = 0$$

$$(iii) \text{ use } d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$d = \sqrt{(20 + 5)^2 + (0 - 20)^2} \quad \boxed{\checkmark}$$

$$\therefore d = \sqrt{1025} = 5\sqrt{41} \quad \boxed{\checkmark}$$

$$(iv) \text{ use } d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

$$\text{use } d = \frac{|4(-3) + 5(4) - 80|}{\sqrt{(4)^2 + (5)^2}} \quad \boxed{\checkmark}$$

$$\therefore d = \frac{72}{\sqrt{41}} \quad \boxed{\checkmark}$$

$$(v) AD = \sqrt{(-4)^2 + 5^2}$$

$$AD = \sqrt{41}$$

$$\text{Area of } ABCD = \frac{1}{2} \times \frac{72}{\sqrt{41}} [\sqrt{41} + 5\sqrt{41}] \quad \boxed{\checkmark}$$

$$= 196 \text{ u}^2 \quad \boxed{\checkmark}$$

Question 4

(a) (i) $\angle SBR = 105^\circ$. ☒ From diagram.

(ii) use $d^2 = 1800^2 + 500^2 - 2 \times 1800 \times 500 \times \cos 105^\circ$

$$\therefore d = \sqrt{1800^2 + 500^2 - 2 \times 1800 \times 500 \times \cos 105^\circ} \quad \text{✓}$$

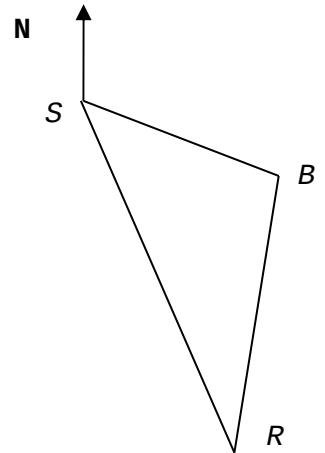
$$d = 1989 \text{ (nearest m)} \quad \text{✓}$$

(iii) now $\angle SRB = \cos^{-1} \left(\frac{1989^2 + 1800^2 - 500^2}{2 \times 1989 \times 1800} \right)$

$$\angle SRB = 14^\circ \quad \text{✓}$$

$\therefore$ Bearing of S from R is

$$(360 - 14^\circ) + 5^\circ = 351^\circ T \quad \text{(nearest degree) ✓}$$



(b) (i) $P(\text{Sarah wins all 3 games}) = 0.9 \times 0.9 \times 0.6$

$$= 0.486 \quad \text{✓}$$

(ii) $P(\text{win 1S, lose 1S and looseB}) + P(\text{win 2S, lose 1S and looseB})$

$$+ P(\text{lose 1S, win 1S and looseB}) \quad \text{✓}$$

$$= 2 \times 0.9 \times 0.1 \times 0.4 + 0.9 \times 0.9 \times 0.4$$

$$= 0.396 \quad \text{✓}$$

(c) 1000, 1250, 1500,, 10000

(i) use $T_n = a + (n-1)d$

$$1000 + (n-1)250 = 10000 \quad \text{✓}$$

$$(n-1)250 = 9000$$

$$(n-1) = \frac{9000}{250}$$

$$\therefore n = 37^{\text{th}} \text{ day} \quad \text{✓}$$

(ii) use $S_n = \frac{n}{2}(a+l)$

$$\therefore S_{37} = \frac{37}{2}(1000 + 10000) \quad \text{✓}$$

$$\therefore S_{37} = 203500m \quad \text{✓}$$

Question 5

(a) $\sin \theta = -\cos \theta$

$$\therefore \tan \theta = -1 \quad (\cos \theta \neq 0) \quad \boxed{\checkmark}$$

$$\therefore \theta = 180 - 45, 360 - 45$$

$$\therefore \theta = 135^\circ, 115^\circ \quad \boxed{\checkmark} \boxed{\checkmark}$$

(b) $5x^2 - 3x - 2 = 0$

(i)
$$\alpha + \beta = \frac{-(-3)}{5}$$
$$= \frac{3}{5} \quad \boxed{\checkmark}$$

(ii)
$$\alpha\beta = \frac{-2}{5} \quad \boxed{\checkmark}$$

(iii)
$$\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta \quad \boxed{\checkmark}$$
$$= \left(\frac{3}{5}\right)^2 - 2\left(\frac{-2}{5}\right)$$
$$= \frac{29}{25} \quad \boxed{\checkmark}$$

(c) (i) since PQRS is a rhombus then $PQ = AP = SP \quad \therefore \Delta APS$ is isosceles $\boxed{\checkmark}$
if $\angle SAP = x^\circ$ then $\angle ASP = x^\circ$
 $\therefore \angle SPQ = 2x$ (exterior angle theorem of a triangle) $\boxed{\checkmark}$

(ii) $\angle SPQ = 2x^\circ$
 $\angle RQP = 180 - 2x$ (co-interior angles as $SP \parallel RQ$) $\boxed{\checkmark}$
since $PQ = QR = QC$ then ΔQRC is an isosceles Δ
 $\therefore \angle QRC = \angle RCQ = 90 - x$ $\boxed{\checkmark}$

since $\angle SAP = x^\circ$ and $\angle RCQ = 90 - x$
then $\Delta ABC = 180 - (90 - x) - x = 90^\circ$ (angle sum of a Δ) $\boxed{\checkmark}$

Question 6(a) *Roots are real and distinct if*

$$> 0 \quad \boxed{\checkmark}$$

$$k^2 - 16 > 0 \quad \boxed{\checkmark}$$

$$\therefore (k - 4)(k + 4) > 0$$

$$\therefore k > 4 \text{ or } k < -4 \quad \boxed{\checkmark}$$

(b) (i) *given* $1800 = Ae^k \dots\dots\dots(1)$

$$\text{and } 2400 = Ae^{2k} \dots\dots\dots(2)$$

 $\therefore (2) \div (1)$ gives

$$e^k = \frac{4}{3} \therefore k = \ln\left(\frac{4}{3}\right) \quad \boxed{\checkmark}$$

$$\therefore Ae^{\ln\left(\frac{4}{3}\right)} = 1800$$

$$\therefore A = \frac{1800}{\cancel{4}/3} = 1350 \quad \boxed{\checkmark}$$

(ii) *let* $N = 4000$

$$\therefore 1350e^{\ln\left(\frac{4}{3}\right)t} = 4000$$

$$\ln\left(\frac{4}{3}\right)t = \ln\left(\frac{4000}{1350}\right)$$

$$\therefore t \approx 3.776 \text{ years} \quad \boxed{\checkmark}$$

$$\text{October 2014} \quad \boxed{\checkmark}$$

$$(iii) \text{ Now } N = 1350e^{\ln\left(\frac{4}{3}\right)t}$$

$$\therefore \frac{dN}{dt} = 1350 \ln\left(\frac{4}{3}\right) e^{\ln\left(\frac{4}{3}\right)t} \quad \boxed{\checkmark}$$

$$\text{let } t = 4 \quad \therefore \frac{dN}{dt} \approx 1227 \text{ million individuals / year} \quad \boxed{\checkmark}$$

(c) Using Simpson's Rule with 5 function values

x	1	2	3	4	5
ln(x+1)	ln2	ln3	ln4	ln5	ln6



$$\therefore \text{Area} \approx \frac{1}{3}((\ln 2 + \ln 6) + 4(\ln 3 + \ln 5) + 2 \ln 4) = 5.36$$

$$= 5.36 u^2$$



$$\text{or Area} \approx \frac{3-1}{6}[\ln 2 + 4 \ln 3 + \ln 4] + \frac{5-3}{6}[\ln 4 + 4 \ln 5 + \ln 6]$$

$$= 5.36 u^2$$

**Question 7**

$$(a) \sum_{n=1}^{\infty} 5\left(\frac{1}{2}\right)^n = 5\left(\frac{1}{2}\right) + 5\left(\frac{1}{2}\right)^2 + 5\left(\frac{1}{2}\right)^3 + \dots \quad \boxed{\checkmark}$$

$$\text{as } S_{\infty} = \frac{a}{1-r}$$

$$S_{\infty} = \frac{\frac{5}{2}}{1 - \frac{1}{2}} = \frac{5}{2-1} = 5 \quad \boxed{\checkmark}$$

$$(b) (i) f(x) = \frac{x^3}{3} + x - 3$$

$$f'(x) = x^2 + 1 \quad \boxed{\checkmark}$$

$$\text{let } f'(x) = 0$$

$$\text{as } x^2 \neq -1 \quad \boxed{\checkmark}$$

$$f'(x) \neq 0 \quad \therefore \text{no stationary points.}$$

$$(ii) f''(x) = 2x$$

$$\text{concave up when } f''(x) > 0$$

$$\therefore \text{when } x > 0 \quad \boxed{\checkmark}$$

(c) (i) $y = \ln x$

$$\frac{dy}{dx} = \frac{1}{x} \quad \therefore \text{Gradient} = \frac{1}{e} \quad \text{when } x = e.$$

$$\therefore \text{tangent is } y - 1 = \frac{1}{e}(x - e) \quad \therefore y = \frac{1}{e}x \quad \boxed{\checkmark}$$

$$(0, 0) \text{ satisfies this} \quad \boxed{\checkmark}$$

(ii) $\frac{d}{dx}(x \ln x - x) = \ln x + x \times \frac{1}{x} - 1 \quad \boxed{\checkmark}$

$$= \ln x + 1 - 1 \quad \boxed{\checkmark}$$

$$= \ln x$$

(iii) $\text{Area} = \text{Area of triangle} - \int_1^e \ln x \, dx$

$$= \frac{1}{2} \times e \times 1 - [x \ln x - x]_1^e \quad \boxed{\checkmark}$$

$$= \frac{e}{2} - (e \times 1 - e + 1) \quad \boxed{\checkmark}$$

$$= \frac{e}{2} - 1 \quad \boxed{\checkmark}$$

Question 8

$$(a) \quad (i) \quad \text{let} \quad \frac{dx}{dt} = e^{3t-2}$$

$$\therefore \frac{d^2x}{dt^2} = 3e^{3t-2}$$

$$\text{when } t = 1 \quad \frac{d^2x}{dt^2} = 3e \quad \boxed{\checkmark}$$

$$(ii) \quad \text{let } e^{3t-2} = 22.3$$

$$\therefore 3t - 2 = \ln 22.3 \quad \boxed{\checkmark}$$

$$t = \frac{2 + \ln 22.3}{3}$$

$$\therefore t = 1.7 \text{ sec} \quad \boxed{\checkmark}$$

$$(iii) \quad \text{dist in 1st sec} = \int_0^1 e^{3t-2} dt$$

$$= \frac{1}{3} [e^{3t-2}]_0^1 \quad \boxed{\checkmark}$$

$$= \frac{1}{3} [e - e^{-2}]$$

$$= 0.861m \quad \boxed{\checkmark}$$

b) The graph of $y = \sin 2x$ from $0 \leq x \leq \pi$ is shown below.

$$\text{let} \quad \int_0^k \sin 2x \, dx = 0.85$$

$$-\frac{1}{2} [\cos 2x]_0^k = 0.85 \quad \boxed{\checkmark}$$

$$\cos 2k - 1 = -1.7$$

$$\cos 2k = -0.7 \quad \boxed{\checkmark}$$

$$2k = \pi - \cos^{-1} 0.7$$

$$k = 1.17 \quad \boxed{\checkmark}$$

(correct to two decimal places)

(c) (i) $P(x, y)$ moves in such a way that PA is perpendicular to PB ,

$$\therefore \frac{y-4}{x} \times \frac{y}{x-2} = -1 \quad \boxed{\checkmark}$$

$$\therefore y^2 - 4y = -x^2 - 2x$$

$$\therefore x^2 + 2x + y^2 - 4y = 0 \quad \boxed{\checkmark}$$

$$(ii) \quad (x+1)^2 + (y-2)^2 = 5$$

$$\text{centre } (-1, 2) \text{ and radius} = \sqrt{5} \quad \boxed{\checkmark} \boxed{\checkmark}$$

Question 9

(a) $y = 3x^2 - 12$

$$\therefore x^2 = \frac{1}{3}(y+12) \quad \boxed{\checkmark}$$

$$\therefore \text{Volume} = \pi \int_0^6 x^2 dy$$

$$V = \pi \int_0^6 \frac{1}{3}(y+12) dy \quad \boxed{\checkmark}$$

$$V = \frac{\pi}{3} \left[\frac{y^2}{2} + 12y \right]_0^6 \quad \boxed{\checkmark}$$

$$V = \frac{\pi}{3}(18 + 72)$$

$$V = 30\pi \text{ cubic units} \quad \boxed{\checkmark}$$

(b) (i) $\text{Arc length} = r\theta = 15 \times \frac{\pi}{6} = \frac{5\pi}{2} \text{ m} \quad \boxed{\checkmark}$

(ii) $\text{Area of sector} = \frac{1}{2}r^2\theta = \frac{225\pi}{12} \text{ m}^2 \quad \boxed{\checkmark}$

(iii)

In $\triangle C1OM$

$$\sin \frac{\pi}{6} = \frac{MC1}{15}$$

$$\therefore MC1 = 15 \times \frac{1}{2} = \frac{15}{2} \quad \boxed{\checkmark}$$

$$\therefore \text{Height} = 15 + \frac{15}{2} = 22.5 \text{ m} \quad \boxed{\checkmark}$$

(iv) $h(t) = 15 - 15 \cos(2t + \frac{\pi}{4})$

$$h'(t) = 30 \sin(2t + \frac{\pi}{4}) \quad \boxed{\checkmark}$$

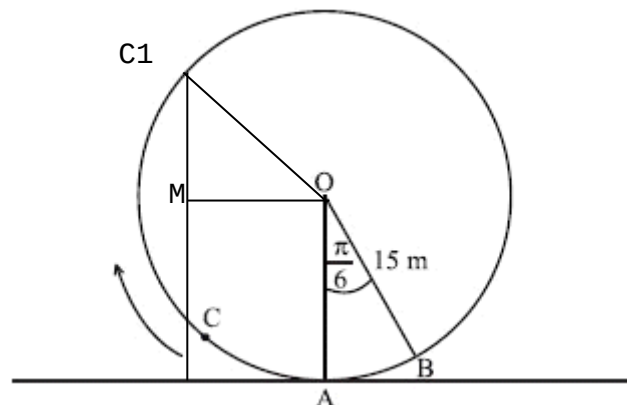
$$h''(t) = 60 \cos(2t + \frac{\pi}{4})$$

let $h''(t) = 0$, height changes most rapidly when acceleration is zero.

$$60 \cos(2t + \frac{\pi}{4}) = 0 \quad \boxed{\checkmark}$$

$$2t + \frac{\pi}{4} = \frac{\pi}{2}$$

$$\therefore t = \frac{\pi}{8} \text{ seconds.} \quad \boxed{\checkmark}$$



Question 10

(a) Let $r = 1.015$ and $\$A_n$ be the amount owing at the end of the n th month.

(i)

$$\therefore A_1 = 35000 - M$$

$$A_2 = 35000 - 2M$$

$$A_3 = 35000 - 3M$$

$$\therefore A_6 = 35000 - 6M. \quad \boxed{\checkmark}$$

$$(ii) \quad A_7 = (35000 - 6M)1.015 - M \quad \boxed{\checkmark}$$

$$A_8 = ((35000 - 6M)1.015 - M)1.015 - M$$

$$\therefore A_8 = (35000 - 6M)1.015^2 - M1.015 - M \quad \boxed{\checkmark}$$

$$\therefore A_8 = (35000 - 6M)1.015^2 - M(1.015 + 1).$$

$$(iii) \quad A_9 = (35000 - 6M)1.015^3 - M(1.015 + 1)1.015 - M$$

$$A_9 = (35000 - 6M)1.015^3 - M(1.015^2 + 1.015 + 1)$$

$$\therefore A_{60} = (35000 - 6M)1.015^{54} - M(1 + 1.015 + 1.015^2 + \dots + 1.015^{53}) \quad \boxed{\checkmark}$$

$$\therefore A_{60} = (35000 - 6M)1.015^{54} - M \left(\frac{1.015^{54} - 1}{1.015 - 1} \right)$$

$$A_{60} = (35000 - 6M)1.015^{54} - M \left(\frac{1.015^{54} - 1}{0.015} \right). \quad \boxed{\checkmark}$$

$$(iv) \quad \text{let } A_{60} = 0$$

$$(35000 - 6M)1.015^{54} - M \left(\frac{1.015^{54} - 1}{0.015} \right) = 0$$

$$M \left(\frac{1.015^{54} - 1}{0.015} \right) = 35000 \times 1.015^{54} - 6 \times 1.015^{54} M$$

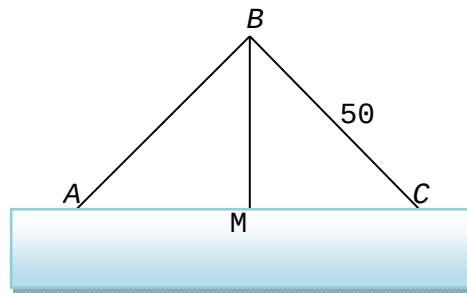
$$M \left(\frac{1.015^{54} - 1}{0.015} \right) + 6 \times 1.015^{54} M = 35000 \times 1.015^{54}$$

$$M \left(\frac{1.015^{54} - 1}{0.015} + 6 \times 1.015^{54} \right) = 35000 \times 1.015^{54}$$

$$M = 35000 \times 1.015^{54} \div \left(\frac{1.015^{54} - 1}{0.015} + 6 \times 1.015^{54} \right) \quad \boxed{\checkmark}$$

$$M = \$817.17 \quad (\text{nearest cent}) \quad \boxed{\checkmark}$$

(b)



(i) as ABC is isosceles. ($AB = BC$)

$$MC = x \quad (BMC \equiv BMA)$$

Using Pythagoras

$$BM^2 = 2500 - x^2$$

$$\therefore BM = \sqrt{2500 - x^2} \quad \boxed{\checkmark}$$

$$\therefore \text{area of } ABC = \frac{1}{2} \times AC \times BM$$

$$\therefore \text{area of } ABC = \frac{1}{2} \times 2x \times \sqrt{2500 - x^2} \quad \boxed{\checkmark}$$

$$\therefore \text{area of } ABC = x\sqrt{2500 - x^2}$$

(ii) Now $A(x) = x(2500 - x^2)^{\frac{1}{2}}$

$$A'(x) = x \times \frac{1}{2} (2500 - x^2)^{-\frac{1}{2}} \times (-2x) + (2500 - x^2)^{\frac{1}{2}} \times 1$$

$$A'(x) = \frac{-x^2}{\sqrt{2500 - x^2}} + \sqrt{2500 - x^2} \quad \boxed{\checkmark}$$

$$A'(x) = \frac{-x^2 + 2500 - x^2}{\sqrt{2500 - x^2}}$$

$$A'(x) = \frac{2500 - 2x^2}{\sqrt{2500 - x^2}}$$

$$\text{let } A'(x) = 0 \quad \therefore 2500 - 2x^2 = 0$$

$$\therefore x = \sqrt{1250} \quad (x > 0)$$

$$\therefore x = 25\sqrt{2} \quad \boxed{\checkmark}$$

First derivative test : $A'(25\sqrt{2} - 0.2) > 0$

$$A'(25\sqrt{2} + 0.2) < 0 \quad \therefore \text{Max when } x = 25\sqrt{2}. \quad \boxed{\checkmark}$$

$\therefore$ Max area occurs when $AC = 50\sqrt{2}$

$AC = 71\text{m}$ (nearest m)